

AP CSP | Lesson 28: Bias & Crowdsourcing

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: B

B allows the same systematic omissions despite increased size. A assumes representation; C invents inclusion; D confuses quantity with accuracy.

Q2. Answer: C

C connects the chosen measure to a misleading ranking. A is unrelated; B overgeneralizes because distinct valid observations may repeat locations; D is not implied by the scenario.

Q3. Answer: A

A obtains input from many people via the Internet. B and C lack crowd input; D concerns computational work distribution rather than human contributions.

Q4. Answer: C

C involves distributed people contributing relevant research data, including non-scientists. A and D do not describe scientific participation; B is not human research contribution.

Q5. Answer: B and D

B describes an evaluation step; D states a bounded mitigation claim. A ignores related signals and data gaps; C ignores inaccurate contributions and selection effects.

Q6. Answer: C

C identifies a remaining participation mechanism. A equates opportunity with action; B concerns formatting rather than selection; D invents identical views.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: Data risk: non-users are missing. Offer accessible alternative contribution routes with the same definitions. Design risk: repeated submissions inflate rankings. Detect duplicate submissions and evaluate whether the ranking better reflects valid evidence.

1 mark: Identifies missing non-users as a data coverage risk.

1 mark: Matches a relevant inclusion action to that coverage gap.

1 mark: Identifies repeated submissions or activity being mistaken for need as a design risk.

1 mark: Matches a rule or validation change to that design risk.

Q8. 4 marks

Answer: A: $(80 \times 0.20 + 20 \times 0.80) / 100 = 32\%$. B: $(40 \times 0.20 + 60 \times 0.80) / 100 = 56\%$. B is 24 percentage points higher. Composition changes while the two group rates stay fixed.

1 mark: Correct sample A rate 32% with calculation.

1 mark: Correct sample B rate 56% with calculation.

1 mark: Correct difference of 24 percentage points.

1 mark: Explains that the sample mix changes while group rates remain fixed.

Q9. 4 marks

Answer: $0.60 \times 0.20 + 0.40 \times 0.80 = 0.44 = 44\%$. It stays fixed while population shares and the two group rates remain fixed. It does not remove within-group nonresponse, self-selection or measurement bias.

1 mark: Uses the correct weighted sum with both population shares.

1 mark: Obtains 44%.

1 mark: States that both population shares and group rates remain fixed.

1 mark: Names a relevant residual bias, such as within-group nonresponse or measurement error.

Q10. 4 marks

Answer: Question: how does shade coverage at sampled public stops change by time of day? Participants submit time-stamped observations using a common protocol on their devices. Many people can cover more locations or times. Audit a sample of observations against independent field checks.

1 mark: States a suitable scientific research question.

1 mark: Specifies relevant data contributed by distributed participants using computing devices.

1 mark: Explains a plausible scale or collaboration benefit.

1 mark: Gives a concrete quality check relevant to the contributed measurements.

Q11. 4 marks

Answer: It can detect missing required fields. It cannot prove the location was actually observed or the report is accurate. An independent field audit can check a sample. The audit may miss errors in unaudited reports.

1 mark: Identifies missing-field detection.

1 mark: Distinguishes complete fields from accurate or genuine observations.

1 mark: Proposes a relevant verification step using independent evidence.

1 mark: States a specific remaining limitation of that step.

Q12. 4 marks

Answer: Crowdsourcing can expand participation but does not guarantee representation or accurate input. The 44% summarizes assumptions, not verified real-world need. Offer supported contribution routes for missing participants. Compare who completes valid submissions before and after, while checking whether important groups remain absent.

1 mark: Corrects the claim that crowdsourcing removes bias.

1 mark: Explains why synthetic weighted 44% does not prove real need.

1 mark: Gives a specific action addressing an identified inclusion barrier.

1 mark: Gives an observable measure that evaluates the action, rather than asserting guaranteed success.